



Lesson 1: Cost of Capital

Module : Valuing Risky Projects

Introduction

In this lesson we will cover the following topics:

- Introduction Company and Project Cost of Capital
- Computing Company Cost of Capital
- Estimating the components of WACC
- Analyzing Project Risk
- Certainty Equivalents
- Summary and concluding remarks



Company and Project Cost of Capital

Company and Project Cost of Capital

- Company cost of capital is defined as the expected return on a portfolio of all the company's existing securities
 - It is the opportunity cost of capital for investment in the firm's assets
 - If the firm has no debt outstanding, then the company cost of capital is just the expected rate of return on the firm's stock
 - The company cost of capital is not the correct discount rate if the new projects are more or less risky than the firm's existing business
 - $Firm\ Value = PV(AB) = PV(A) + PV(B)$
 - The two discount rates will, in general, be different

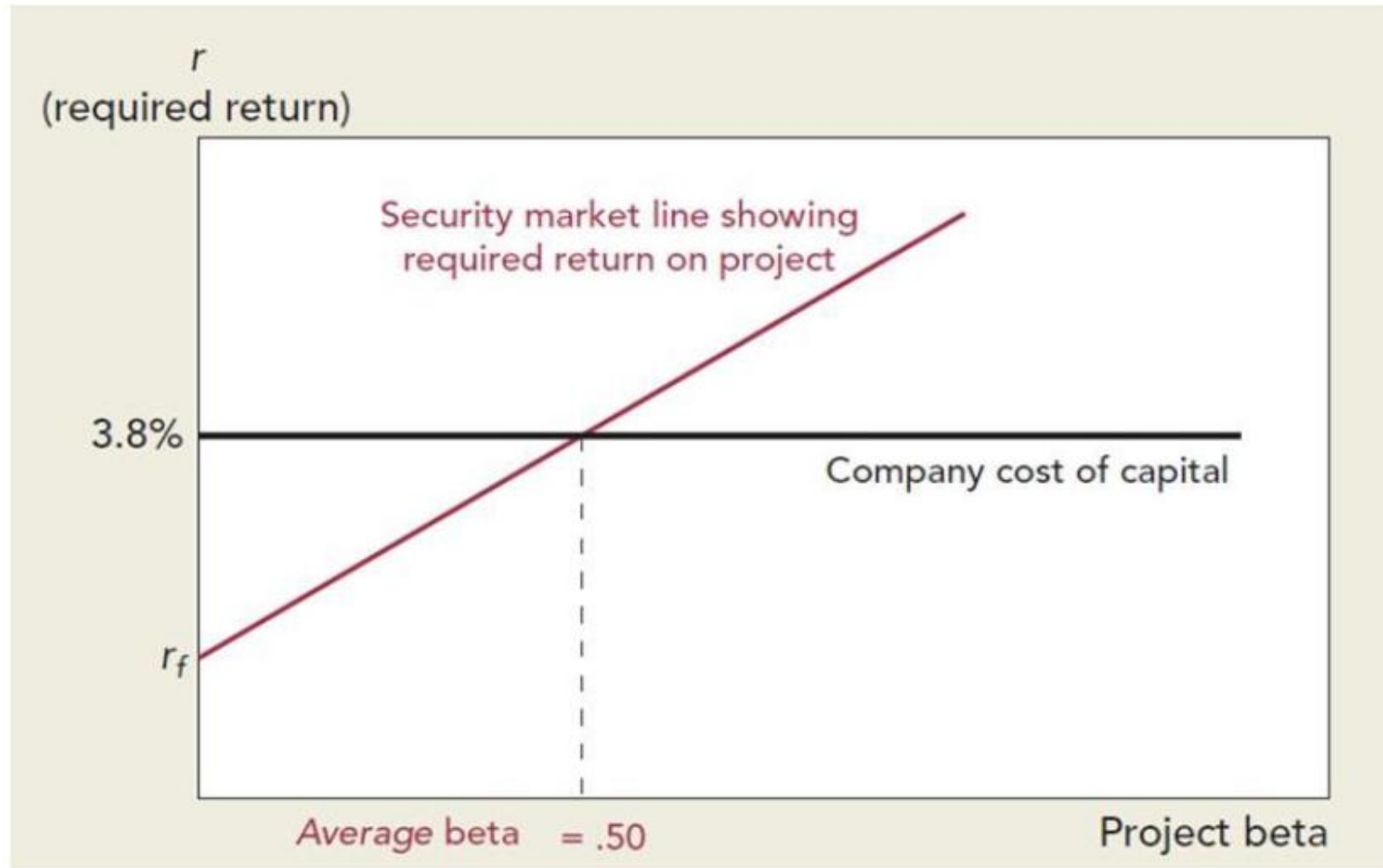
Company and Project Cost of Capital

- If the present value of an asset depended on the identity of the company that bought it, present values would not add up
 - Consider a portfolio of \$1 million invested in firm A and \$1 million invested in firm B
 - If the firm considers investing in a third project C, it should also value C as if C were a mini-firm
 - The opportunity cost of capital depends on the use to which that capital is put
 - A new division with a different risk profile, considerable uncertainty and customer demand that is yet to be established, should of course have different cost of capital

Company and Project Cost of Capital

- Suppose we measure the risk of each project by its beta
 - Then a firm should accept any project lying above the upward-sloping security market line that links expected return to risk
 - If the project is high-risk, the firm needs a higher prospective return than if the project is low-risk
 - That is different from the company cost of capital rule, which accepts any project regardless of its risk as long as it offers a higher return than the company's cost of capital

Company and Project Cost of Capital



Company and Project Cost of Capital

- The true cost of capital depends on project risk, not on the company undertaking the project
 - Why is so much time spent estimating the company cost of capital?
 - First, many (maybe most) projects can be treated as average risk
 - Second, the company cost of capital is a useful starting point for setting discount rates for unusually risky or safe projects
 - It is easier to add to, or subtract from, the company cost of capital

Company and Project Cost of Capital

- Businesspeople have good intuition about relative risks
 - They set a companywide cost of capital as a benchmark
 - Many large companies use the company cost of capital not just as a benchmark, but also as an all-purpose discount rate for every project proposal
 - Measuring differences in risk is difficult to do objectively
 - Top management may demand extra- conservative cash-flow forecasts from extra-risky projects



Computing Company Cost of Capital

Computing Company Cost of Capital

- Company cost of capital is the expected return on a portfolio of all the company's existing securities
 - The portfolio of company security usually includes debt as well as equity
 - Thus the cost of capital is estimated as a blend of the cost of debt

Asset value	100	Debt	$D = 30$ at 7.5%
		Equity	$E = 70$ at 15%
Asset value	100	Firm value	$V = 100$

- The values of debt and equity add up to overall firm value

Computing Company Cost of Capital

- The company cost of capital is not equal to the cost of debt or to the cost of equity but is a blend of the two
 - Suppose you purchased a portfolio consisting of 100% of the firm's debt and 100% of its equity
 - The expected rate of return on your hypothetical portfolio is the company cost of capital
 - The expected rate of return is just a weighted average of the cost of debt ($r_D = 7.5\%$) and the cost of equity ($r_E = 15\%$)
 - The weights are the relative market values of the firm's debt and equity, that is, $D/V = 30\%$ and $E/V = 70\%$

Computing Company Cost of Capital

- The company cost of capital is not equal to the cost of debt or to the cost of equity but is a blend of the two
 - The marginal corporate tax rate $T_c = 35\%$
 - WACC or Company cost of capital = $r_D * (1 - T_c) * \frac{D}{V} + r_E * \frac{E}{V} = 7.5\% * (1 - 0.35) * 0.30 + 15 * 0.70 = 12.00\%$
 - This blended measure of the company cost of capital is called the weighted-average cost of capital or WACC



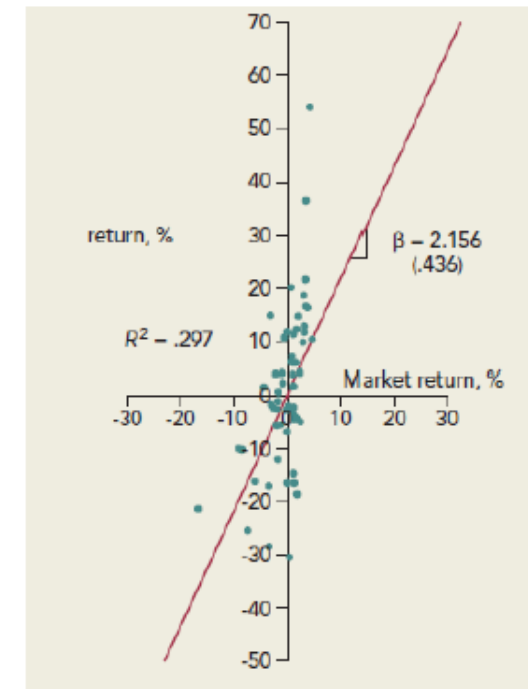
Estimating the components of WACC

Estimating the components of WACC

- To calculate the weighted-average cost of capital, you need an estimate of the cost of equity
 - We will use the capital asset pricing model (CAPM) to estimate the cost of equity
 - CAPM: *Expected Returns* $= r_f + \beta(r_m - r_f)$
 - In principle, we are interested in the future beta of the company's stock
 - We will estimate beta using historical security price data

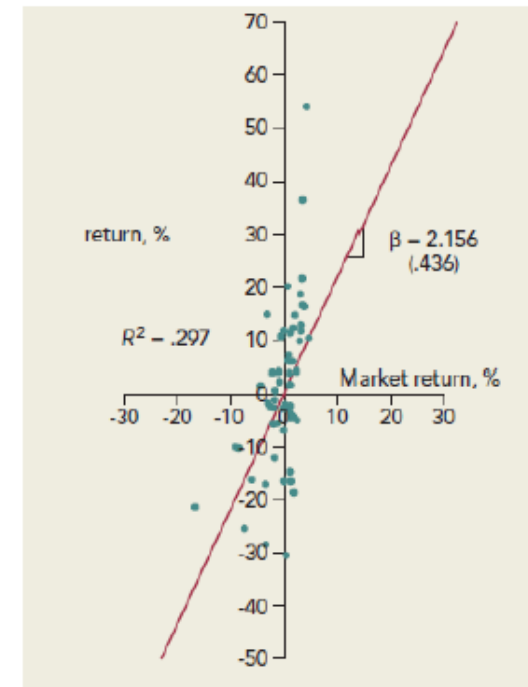
Estimating the components of WACC

- In the scatter diagram shown here, each dot represents the return on a security and return on market
 - The slope of fitted line is called beta
 - The R-square measure tells the proportion of total variance that can be explained by market variance
 - It appears that 29.7% of the variance was explained by the market
 - The 95% confidence interval estimate of beta is 2.16



Estimating the components of WACC

- In the scatter diagram shown here, each dot represents the return on a security and return on market
 - Standard error of the estimated beta is computed to show the extent of possible mismeasurement
 - Standard error of beta estimate is 0.436
 - The 95% confidence interval estimate 2.16 plus or minus 2×0.436
 - That is, you have 95% chance of being right in saying that beta can fall in this interval



Estimating the components of WACC

- How to estimate the risk-free rate of interest (r_f)
 - Should we use short-term treasury bill rate, daily over night rate, monthly rate, one year interest rate or long-term interest rates
 - CAPM is a short-term model
 - It works period by period and calls for a short-term interest rate
 - Could a .2% three-month risk-free rate give the right discount rate for cash flows 10 or 20 years in the future?

Estimating the components of WACC

- How to estimate the risk-free rate of interest (r_f)
 - Financial managers can simply use a long-term risk-free rate in the CAPM formula
 - Suppose the risk-free rate is 1.8% and the market risk-premium is 7%, and beta is 1.16, then the cost of equity can be computed as follows.
$$\text{Cost of equity} = \text{Expected Returns} = r_f + \beta(r_m - r_f) = 1.8 + 1.16 * 7.0 = 9.9\%$$

Estimating the components of WACC

- Estimating cost of equity and WACC
 - Let us calculate the WACC for a firm with cost of Debt of about 7.8%, corporate tax-rate of 35%, and debt ratio (D/V) of 31.5%.
 - *After Tax WACC* $= (1 - T_C) * r_D * \frac{D}{V} + r_E * \frac{E}{V} = (1 - 0.35) * 7.8 * 0.315 + 9.9 * 0.685 = 8.4\%$

Estimating the components of WACC

- The cost of debt is always less than the cost of equity
 - The WACC formula blends the two costs: Debt and Equity
 - As the debt ratio D/V increases, the cost of the remaining equity also increases
 - This offsets offsetting the apparent advantage of more cheap debt
 - Debt does have a tax advantage, however, because interest is a tax-deductible expense

Thanks!